

Microsoft Purview – Data Governance & Cataloguing

Azure Data Governance Hackathon Project | B.Tech 3rd-Year Implementation Plan

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Domain: Data Governance | **Platform:** Microsoft Azure | **Core Service:** Microsoft Purview

1. Abstract

Modern organizations store data across databases, cloud storage, analytics platforms, and collaboration systems. When data is spread across these sources, teams may struggle to discover where data exists, understand its sensitivity, trace its movement, and apply consistent governance rules. This project proposes a Microsoft Purview-based data governance and cataloguing solution that scans five representative data sources—Azure SQL Database, Azure Blob Storage, Azure Data Lake Storage Gen2, Azure Synapse Analytics, and SharePoint Online. The solution uses Microsoft Purview Data Map and Catalog capabilities to collect metadata, classify sensitive information, support data discovery, and provide governance visibility. The design also includes lineage, access-policy awareness, monitoring, and audit capabilities. The project is intentionally designed as a practical B.Tech-level prototype that can be demonstrated during a hackathon using sample datasets and controlled credentials rather than production data.

2. Problem Statement

The team needs a single governance view across multiple data sources. Without a catalogue, users may not know which dataset is authoritative, where sensitive information is stored, or how data moves between systems. Manual documentation also becomes outdated quickly. The proposed solution centralizes metadata discovery through Microsoft Purview, enabling searchable data assets, classification of sensitive information, lineage visibility where supported, and governance-oriented monitoring.

3. Objectives

- Catalogue and classify data across five representative sources.
- Identify locations containing sensitive information such as names, emails, phone numbers, IDs, and financial fields.
- Provide a searchable inventory of datasets and their metadata.
- Show data lineage for supported processing and analytics flows.
- Create a simple governance model for ownership, access, sensitivity, and auditability.
- Demonstrate the solution with sample data so the project can be completed without production credentials.

4. Proposed Architecture

Five representative sources feed metadata into Microsoft Purview, which provides discovery, cataloguing, classification and governance visibility.

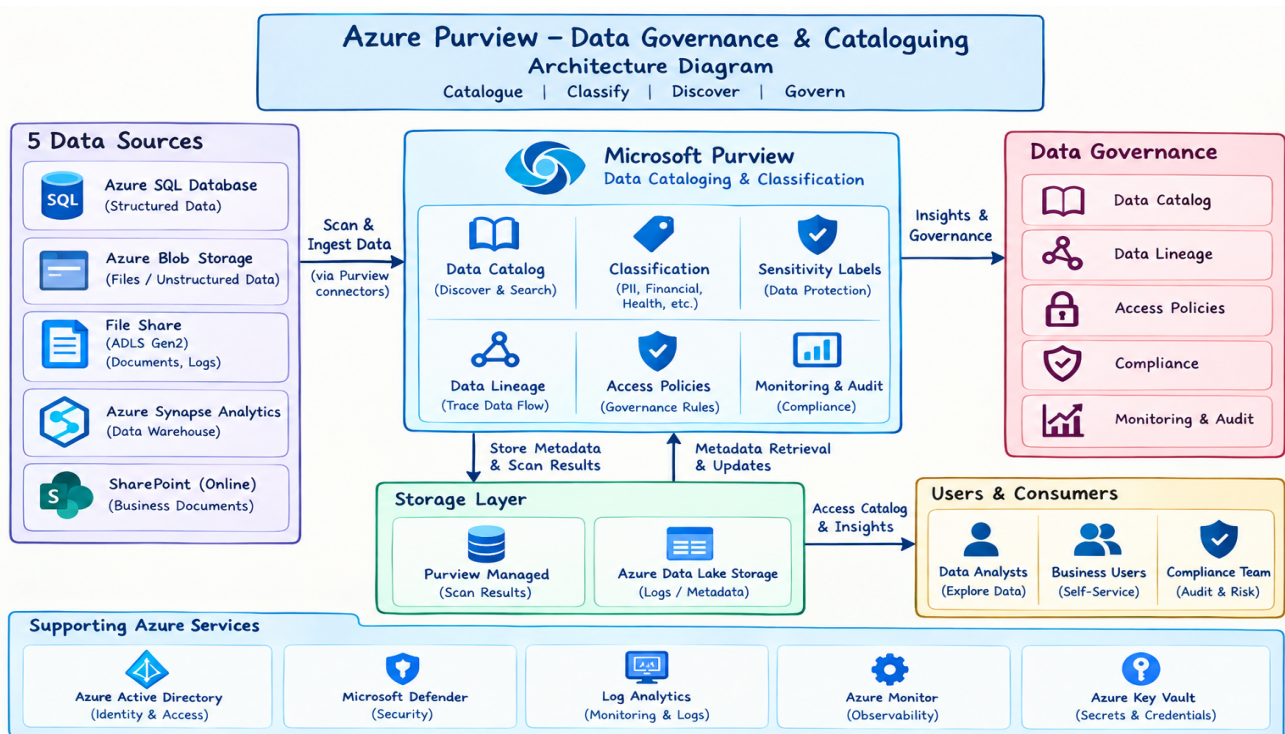


Figure 1. Microsoft Purview – Data Governance & Cataloguing Architecture

5. Data Sources

Data Source	Role	Sample Content
Azure SQL Database	Structured transactional data	Customer/user tables, orders, employee records
Azure Blob Storage	Files and unstructured/object data	CSV, JSON, text files, sample documents
Azure Data Lake Storage Gen2	Data lake / files	Raw and curated datasets, logs
Azure Synapse Analytics	Analytics / warehouse	Curated tables and reporting datasets
SharePoint Online	Business documents	Policies, reports, project documents

6. Azure Services / Technologies

Service	Purpose	Project Use
Microsoft Purview	Core governance platform	Data Map, cataloguing, scanning, classification, discovery
Microsoft Purview Unified Catalog / Data Catalog	Search and discovery	Find data assets, owners, glossary/metadata and governance context
Azure SQL Database	Data source	Structured sample data for scanning and classification
Azure Blob Storage	Data source	Files/object data for metadata discovery
Azure Data Lake Storage Gen2	Data source	Raw/curated data lake files
Azure Synapse Analytics	Data source / analytics	Warehouse or analytics datasets and lineage scenarios
SharePoint Online	Data source	Business documents and collaboration data
Microsoft Entra ID	Identity	Authentication, RBAC and controlled access
Azure Key Vault	Secrets	Store credentials/secrets where required by the implementation
Azure Monitor + Log Analytics	Monitoring	Operational visibility, scan/solution monitoring and troubleshooting

7. Implementation Plan

1. Prepare small, synthetic datasets in each source.
2. Create/configure Microsoft Purview and organize collections/domains for the project.
3. Register data sources and configure the required connections/credentials.
4. Run scans using suitable scan rule sets.
5. Review discovered assets, schemas, classifications and metadata.
6. Validate sensitive-data classifications with a team member acting as the business/data owner.
7. Document ownership, sensitivity, access requirements and governance rules.
8. Demonstrate search/discovery and lineage for supported sources and pipelines.
9. Capture evidence screenshots and metrics for the hackathon presentation.

8. Bottlenecks and Mitigation

Bottleneck	Mitigation
Credentials unavailable	Use synthetic data and a dedicated student/demo resource group; request only minimum required roles.
Classification needs business validation	Create a validation checklist and manually confirm a sample of classifications before the final demo.
Connector/licensing differences	Record the exact connector/configuration available in the tenant and keep a CSV/JSON sample-data fallback.
Lineage not visible for every source	Demonstrate lineage only for supported flows and clearly label unsupported/limited cases.
Sensitive data exposure during demo	Use synthetic PII-like values; never upload real personal or confidential information.
Scan failures or timeouts	Start with small datasets, verify network/authentication settings, then increase scan scope.

9. Hackathon MVP Acceptance Criteria

- Five registered data sources with successful scans where connector access is available.
- A searchable catalogue containing representative assets.
- Classification results for at least 3–5 sensitive data types.
- A documented sensitivity taxonomy: Public, Internal, Confidential, Restricted.
- At least one end-to-end lineage demonstration for a supported source/process.
- Screenshots/evidence of scan status, catalog search, classification and governance view.

10. Team Demo Pitch

Our project treats data governance as a discoverability and trust problem. Instead of maintaining separate spreadsheets for every database, file store and business document, we use Microsoft Purview as a central governance layer. We scan five representative sources, automatically discover metadata, classify sensitive information, and expose governance context to data analysts, business users and compliance teams. The prototype is safe for a hackathon because it uses synthetic data, minimum-access credentials, and measurable evidence such as scan success, asset count, classification coverage and lineage visibility.

11. GitHub Structure

```
azure-purview-data-governance/
├── README.md
├── docs/
│   ├── ABSTRACT.md
│   ├── ARCHITECTURE.md
│   ├── SERVICES.md
│   ├── IMPLEMENTATION.md
│   ├── BOTTLENECKS.md
│   └── DEMO-SCRIPT.md
├── architecture/
│   └── purview-data-governance-architecture.png
├── sample-data/
│   ├── sql/
│   ├── blob/
│   ├── adls-gen2/
│   └── synapse/
├── evidence/
│   └── screenshots/
└── LICENSE
```

12. Conclusion

The project provides a practical, safe and demonstrable governance prototype. The team can show how Microsoft Purview discovers and classifies data across multiple sources, while the supporting Azure services provide identity, secrets and monitoring foundations.

Official references: Microsoft Purview overview; Data governance; classification; lineage; sensitivity labels.